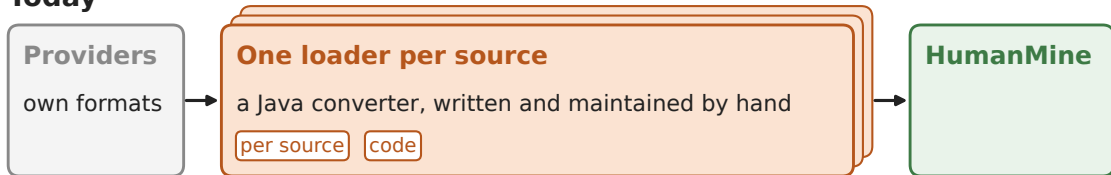


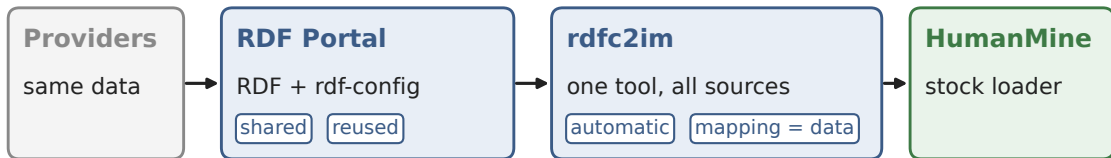
RDF Portal + rdf-config are enough to rebuild an InterMine

The mapping is data, not code. (HumanMine: no data release since February 2022.)

Today



With RDF Portal and rdfc2im



Building InterMine databases from RDF Portal • Russell Y. Neches (Kyoto Univ.), Shuichi Kawashima, Toshiaki Katayama

(NIG / DBCLS), Gos Micklem (Univ. of Cambridge) • DBCLS BioHackathon 2026

It works: a HumanMine built from RDF Portal

9 sources, 8 of them
via RDF Portal

113 panel genes: food
and drug
metabolism

193,288 genes from NCBI
Gene, in full

53,105 ClinVar alleles for
the panel

search • templates • list analysis

Filter by text
Search for keywords

Filter by category
All Genomics Proteins Literature GWAS Gene Ontology Disease Variant

GWAS associations for a gene

Show GWAS Catalog associations for a gene, with phenotype, p-value and the associated SNP.

Gene > Symbol
= *

Results Preview

Associated Genes > Symbol	GWAS Result > Phenotype	GWAS Result > P Value	GWAS Result > SNP . Primary Identifier
ABCB1	age at menarche, spine bone mineral density	8e-7	rs10264856
ABCB1	age at menarche, spine bone mineral density	8e-7	rs10264856
ABCB1	age at menarche, spine bone mineral density	0.000002	rs2157926
ABCB1	age at menarche, spine bone mineral density	0.000002	rs2157926
ABCB1	alpha wave measurement	0.000001	rs12720086

+ 27502 more results

VIEW 27507 ROWS RESET EDIT QUERY

Web service URL

Close <<

Categories: Genomics GWAS

The running mine: a GWAS Catalog template added for the panel.

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search • templates • list analysis

The screenshot shows the HumanMine interface with a search bar and filters. The main table displays GWAS associations for a specific gene. The table has four columns: Gene, GWAS Result > Phenotype, GWAS Result > P Value, and GWAS Result > SNP - Primary Identifier. The data rows show associations for the gene ABCB1 with the phenotype 'age at menarche, spine bone mineral density' and the SNP 'rs10264856'. The p-value is 8e-7. The table is overlaid with a large red stamp containing the Japanese characters '共有財産' (kyōyū zaisan), which translates to 'shared asset'.

Gene	GWAS Result > Phenotype	GWAS Result > P Value	GWAS Result > SNP - Primary Identifier
ABCB1	age at menarche, spine bone mineral density	8e-7	rs10264856
ABCB1	age at menarche, spine bone mineral density	8e-7	rs10264856
ABCB1	age at menarche, spine bone mineral density	0.000002	rs2157926
ABCB1	age at menarche, spine bone mineral density	0.000002	rs2157926
ABCB1	alpha wave measurement	0.000001	rs12720086

kyōyū zaisan — shared asset

The hard part is identity, not mapping

- ▶ Which key does each object merge on?
- ▶ A gene needs its organism, as well as its identifier, to merge
- ▶ Some upstream keys are not unique: 255 Ensembl ids are claimed by more than one NCBI gene

For rdf-config: mark which predicate identifies a record.

Next: script a full rebuild of the mine for each RDF Portal release.

github.com/intermineorg/rdfc2im

github.com/ryneches/rdfc2im-report

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